

Linear Algebra – MAT 2610

Section 1.8 (Introduction to Linear Transformations)

Dr. Jay Adamsson

jay@aorweb.ca

jadamsson@upei.ca



Definitions

Definition: A function (or transformation) $f: A \rightarrow B$ is a rule that assigns every element in a set A to an element in set B . (Note that A and B are not matrices here, but sets).

Set A is called the **domain** of f and B is called the **co-domain** of f

Definition: The **range** of a function $f: A \rightarrow B$ is the set of all possible values $f(a)$ where $a \in A$

Example

Suppose $A = \{1, 2, 3, 4, 5\}$ and $B = \{\text{red}, \text{green}, \text{blue}\}$ and $f: A \rightarrow B$ is defined by:

$$f(n) = \begin{cases} \text{red} & \text{if } n \text{ is odd} \\ \text{blue} & \text{if } n \text{ is even} \end{cases}$$

Note that we can define $f(n)$ for any n in A (such as $f(2)$ or $f(5)$). But not every element of B is obtained.

Domain of $f: \{1, 2, 3, 4, 5\}$

Co-Domain of $f: \{\text{red}, \text{green}, \text{blue}\}$

Range of $f: \{\text{red}, \text{blue}\}$



We can define functions $f: S_1 \rightarrow S_2$ where S_1 and S_2 are sets of vectors.

Example: Suppose $f: R_2 \rightarrow R_2$ is defined as $f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3x + y \\ x - 2y \end{bmatrix}$

$$\begin{aligned} \text{Find } f\left(\begin{bmatrix} 2 \\ -1 \end{bmatrix}\right) &= \begin{bmatrix} 3(2) + (-1) \\ (2) - 2(-1) \end{bmatrix} \\ &= \begin{bmatrix} 6 - 1 \\ 2 + 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix} \end{aligned}$$



Example: Suppose $f: R_5 \rightarrow R_3$ is defined as $f\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 + x_2 \\ x_3 + x_4 \\ x_2 - x_3 + 2x_5 \end{bmatrix}$

$$\text{Find } f\left(\begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} 3(1) + 2 \\ (-3) + 4 \\ 2 - (-3) + 2(-5) \end{bmatrix} = \begin{bmatrix} 3 + 2 \\ -3 + 4 \\ 2 + 3 - 10 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -5 \end{bmatrix}$$



Matrix Transformation

If A is an $m \times n$ matrix, then if x is any vector in R^n , let

$$\boxed{T(x)} = \boxed{Ax}$$

Observe that this takes any vector in R^n and “transforms” it into a vector in R^m . This is a **matrix transformation** denoted by $x \mapsto Ax$



Matrix Transformation Example

$T: \underline{\underline{R^3}} \rightarrow \underline{\underline{R^2}}$ defined by $T\left(\underline{\underline{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}}\right) = \underbrace{\begin{bmatrix} 2 & 2 & -3 \\ 3 & -1 & 2 \end{bmatrix}}_{\substack{\text{2x3 matrix} \\ \downarrow}} \underline{\underline{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}}$

eg: $T\left(\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 2 & 2 & -3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

$$= (1) \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 1 \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$
$$= \begin{bmatrix} 2+4+3 \\ 3-2-2 \end{bmatrix} = \begin{bmatrix} 9 \\ -1 \end{bmatrix}$$



Matrix Transformation Example

Is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ in the range of $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 & 2 \\ 3 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

in other words, can I find a vector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
where $T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 & 2 \\ 3 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 + 2x_2 \\ 3x_1 - x_2 \\ 2x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$



$$\begin{bmatrix} 2 & 2 & 1 \\ 3 & -1 & 2 \\ 2 & 0 & 3 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 2 & 0 & 3 \\ 3 & -1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \xrightarrow{3r_1 - 2r_2 \rightarrow r_2} \begin{bmatrix} 2 & 0 & 3 \\ 0 & 2 & 5 \\ 2 & 2 & 1 \end{bmatrix}$$

$$\xrightarrow{r_1 - r_3 \rightarrow r_1} \begin{bmatrix} 2 & 0 & 3 \\ 0 & 2 & 5 \\ 0 & -2 & 2 \end{bmatrix} \xrightarrow{r_3 \leftrightarrow r_2} \begin{bmatrix} 2 & 0 & 3 \\ 0 & -2 & 2 \\ 0 & 2 & 5 \end{bmatrix} \xrightarrow{r_2 + r_3 \rightarrow r_3} \begin{bmatrix} 2 & 0 & 3 \\ 0 & -2 & 2 \\ 0 & 0 & 7 \end{bmatrix}$$

No Solution

$\Rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is not in range of T



Linear Transformation

Definition: A linear transformation $T: R^n \rightarrow R^m$ is a function where the following holds for any vectors $u, v \in R^n$ and c any scalar:

1. $T(u + v) = T(u) + T(v)$ 
2. $T(cu) = cT(u)$ 

Note that every matrix transformation is a linear transformation.



Example: Is $T: R^3 \rightarrow R^2$ defined by $T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{bmatrix} 2 & 2 & -3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ a linear transformation?

$$1) \underline{T(u+v) = T(u) + T(v)}$$

$$\text{Let } u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\begin{aligned} T \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \right) &= T \left(\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2 & 2 & -3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix} = (u_1 + v_1) \begin{bmatrix} 2 \\ 3 \end{bmatrix} + (u_2 + v_2) \begin{bmatrix} 2 \\ -1 \end{bmatrix} + (u_3 + v_3) \begin{bmatrix} -3 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2u_1 + 2v_1 + 2u_2 + 2v_2 - 3u_3 - 3v_3 \\ 3u_1 + 3v_1 - u_2 - v_2 + 2u_3 + 2v_3 \end{bmatrix} \end{aligned}$$



$$\begin{aligned}
T(u) + T(v) &= T\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) + T\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) \\
&= \begin{bmatrix} 2 & 2 & -3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + \begin{bmatrix} 2 & 2 & -3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \\
&= u_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + u_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + u_3 \begin{bmatrix} -3 \\ 2 \end{bmatrix} + v_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + v_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + v_3 \begin{bmatrix} -3 \\ 2 \end{bmatrix} \\
&= \begin{bmatrix} 2u_1 + 2v_1 + 2u_2 + 2v_2 - 3u_3 - 3v_3 \\ 3u_1 + 3v_1 - u_2 - v_2 + 2u_3 + 2v_3 \end{bmatrix}
\end{aligned}$$

This shows $T(u+v) = T(u) + T(v)$ ✓

(2) Show $T(cu) = c(T(u))$ ← Exercise



For any linear transformation $T: R^n \rightarrow R^m$ the following are true:

1. $T(0) = 0$ $\leftarrow T(u-u) = T(u) - T(u) \Rightarrow T(0) = 0$
2. $T(-u) = -T(u)$ for all $u \in R^n$ $\leftarrow$
3. $T(u-v) = T(u) - T(v)$ for all $u, v \in R^n$ $\leftarrow$
4. $T(au + bv) = aT(u) + bT(v)$ for all $u, v \in R^n$ and scalars a, b $\leftarrow$

Note: this means that

$$\begin{aligned} T(a_1 u_1 + \cdots + a_n u_n) &= a_1 T(u_1) + \cdots + a_n T(u_n) \\ &= T(a_1 u_1) + \cdots + T(a_n u_n) \end{aligned}$$



Linear Algebra – MAT 2610

Section 1.8 (Introduction to Linear Transformations)

Dr. Jay Adamsson

jay@aorweb.ca

jadamsson@upei.ca

